103. If a mouse of over eight times its normal size has been produced by introducing human growth hormone gene, the technique involved is called:

(a) Hybridisation (b) Genetic engineering
(c) Mutation Breeding (d) Hormonal feeding

Ans. (b) : If a mouse of over eight times its normal size has been produced by introducing human growth hormone gene, the technique involved is Genetic engineering.

Genetic engineering is the process of using recombinant DNA (rDNA) technology to alter the genetic makeup of an organism. Genetic engineering involves the direct manipulation of one or more genes. Most often, a gene from another species is added to an organism's genome to give it a desired phenotype.

104. Match List I with List II and select the correct answer using the codes given below the lists:

List I (Characters tics)	List II (Animal)
I. Wingless insect	(A) Kiwi
II. Flightless bird	(B) Silver fish
III. Limbless reptile	(C) Turtle
IV. Limbless animal	(D) Snake
	(E) Fish

Codes:

(a) IA, IIC, IIIB, IVE (b) IB, IIA, IIID, IVE
(c) IB, IIA, IIIC, IVD (d) IC, IIA, IIID, IVB

Ans. (b) : The correct match for list 1 and list 2 is

Wingless insect	–	Silver fish
Flightless bird	–	Kiwi
Limbless reptile	–	Snake
Limbless animal	–	Fish

105. Three communicable diseases prevalent in developing countries caused by unsafe drinking water and bad sanitation are:

(a) acutediarrhoea, cancer and gout
(b) Malaria, acute diarrhea and schistosomiasis
(c) onchocerciasis, leukaemia and arthritis
(d) Rheumatism, malaria and AIDS

Ans. (b) : Three communicable diseases prevalent in developing Countries caused by unsafe drinking water and bad sanitation are malaria, acute diarrhea and schistosomiasis.

106. Scratching eases itching because:

(a) It removes the outer dust in the skin.
(b) It kills the germs.
(c) It stimulates certain nerves which direct the brain to increase the production of antihistaminic chemicals.
(d) It suppresses the production of enzymes which cause itching.

Ans. (c) : Scratching eases itching because it stimulates certain nerves which direct the brain to increase the production of antihistaminic chemicals.

107. The blood pressure values of four persons are given below:

1. Mrs. X : 90/60 2. Mr. X : 160/120
3. Mr. Y : 120/80 4. Mrs. Y : 140/100

Who among the following has normal blood pressure?

(a) Mrs. X (b) Mr. X
(c) Mrs. Y (d) Mr. Y

Ans. (d) : Blood pressure is the force of your blood pushing against the walls of your arteries. Each time your heart beats, it pumps blood into the arteries. Your blood pressure is highest when your heart beats, pumping the blood. This is called systolic pressure. When your heart is at rest, between beats, your blood pressure falls. This is called diastolic pressure.

The normal blood pressure is in range of 120/80.Thus Mr. Y has normal blood pressure.

108. Sucrose content in sugarcane decreases:

(a) If high rainfall occurs during the period of growth of the plant.
(b) If frost occurs during the period of ripening.
(c) If there is fluctuation in temperature during the period of growth of the plant.
(d) If there is high temperature during the time of ripening.

Ans. (a) : Sugarcane ripening is the process of sucrose accumulation in stalks, which is heavily influenced by several factors, mainly by climatic conditions such as air temperature and water deficits.

So, Sucrose content in sugarcane decreases If high rainfall occurs during the period of growth of the plant.

109. Sugarcane, sugarbeet, sweetpea, chickpea, pigeonpea and French bean belong to:

(a) Two plant families
(b) Three plant families
(c) Four plant families
(d) Five plant families

Ans. (b) : Sugarcane, sugarbeet, sweetpea, chickpea, pigeonpea and French bean belong to three plant family.

110. Biofilms are:

(a) colour films used by biologists to photograph living organisms.
(b) photomicrographs of microorganisms from various habitats.
(c) accumulations of microorganisms at surfaces such as those of a rock, a tooth or an oil droplet.
(d) photographic films made through fermentation processes of certain industrial microorganisms.

Ans. (c) : Bio films are accumulations of microorganisms at surfaces such as those of a rock, a tooth or an oil droplet.

A bio film forms when certain microorganisms (for example, some types of bacteria) adhere to the surface of some object in a moist environment and begin to reproduce. The microorganisms form an attachment to the surface of the object by secreting a slimy, glue-like substance.

111. Which one of the following approaches comes under the category of biotechnology ?

(a) Use of living organisms or substances obtained from them in industrial processes.
(b) Modernizing the commercial industrial to produce products for use in biological research.
(c) Use of modern technology to investigate biological disorders
(d) Use of industrial technology to increase the biosphere

Ans. (a) : Use of living organisms or substances obtained from them in industrial processes.

ENVIRONMENT AND ECOLOGY

112. Which of the following strongly threatens biodiversity?

(a) Fragile ecosystems such as mangroves and wetlands.
(b) Inaccessible habitats in Himalayas.
(c) Destruction of natural habitats and vegetation and shifting cultivation.
(d) Creation of biosphere reserves.

Ans. (c) : Biodiversity is the biological variety and variability of life on Earth. Biodiversity is a measure of variation at the genetic, species, and ecosystem level. Terrestrial biodiversity is usually greater near the equator, which is the result of the warm climate and high primary productivity.

Major threats to biodiversity are commonly recognized in the programmes of work of the Convention: invasive alien species, climate change, nutrient loading and pollution, habitat change, and over exploitation. Destruction of natural habitats and vegetation and shifting cultivation strongly threatened the biodiversity.

113. Which one of the following does provide the best estimate of world's biological diversity?

(a) Of about ten million species probably alive today, some 20 species are lost everyday, most of them unknown because no more than half a million have yet been actually identified by scientists.
(b) Of about thirty million living species, some 50 are lost everyday, actually identified.
(c) Of about forty five million living species, some 100 are lost every day, most of them unknown because no more than 1.5 million have been actually identified.
(d) Of about seventy five million living species, some 500 are lost every day, most of them unknown because no more than 3 million have been actually identified.

Ans. (c) : The term biodiversity (from "biological diversity") refers to the variety of life on Earth at all its levels, from genes to ecosystems, and can encompass the evolutionary, ecological, and cultural processes that sustain life.

The best estimate of world's biological diversity of about forty five million living species, some 100 are lost everyday, most of them unknown because no more than 1.5 million have been actually identified.

114. Which one of the following is a man made" cereal NOT found in nature?

(a) Dwarf wheat (b) Hybrid maize
(c) Triticale (d) Soya bean

Ans. (c) : 'Triticale' was prepared by mixing wheat and rye. Man-made artificial grain. It is not available in nature. 'Triticale' is the first artificial grain prepared by scientists.

115. In many developing countries, three major causes of death and disease are:

(a) Lack of safe drinking water, increasing use of pesticides and thinning of the ozone layer.
(b) Contaminated food, global warming and industrial chlorofluorocarbons.
(c) Polluted air, greenhouse effect and soil erosion.
(d) Dirty water, contaminated food and polluted air.

Ans. (d) : In many developing countries of the world which are also called 'third world countries'. The three main causes of death and disease of people in these countries are dirty drinking water and contaminated food and polluted air.

CURRENT AFFAIRS

116. The book The Man who knew Infinity by Robert Kanigel, published in 1991, and extensively reviewed in newspapers in India is the biography of:

(a) Vikram Sarabhai
(b) Homi Bhabha
(c) Sir C.V. Raman
(d) Srinivasa Ramanujan

Ans. (d) : The life character of Indian scientist and mathematician 'Srinivasa Ramanujan' has been mentioned in the book 'The Man who knew Infinity' written by Robert Kanigel in 1991.

117. Which one of the following was part of the erstwhile Soviet Union over which Kurdish population is spread?

(a) Armenia (b) Azerbaijan
(c) Georgia (d) Turkmenistan

Ans. (a) : Armenia was part of the erstwhile soviet union over which Kurdish population is spreaded.

118. Which one of the following monuments figured in the news recently in connection with a fall of some of its parts and its consequent archaeological conservation?

(a) Jagannath Temple, Puri
(b) Sun Temple, Konark
(c) Khandariya Mahadeva Temple, Khajuraho
(d) Chaunsat Yogini Temple, Bheraghat